Sahaj Bhadja

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Summary

Platform software engineer with 2+ years building **Java/Python systems**, scalable storage, reusable APIs, and AI-enabled applications; shipped reliable infrastructure handling 2M+ records and 10K+ daily requests with graceful failure isolation and tested CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Designed **multi-tenant storage systems and reusable data interfaces** for 500+ users, implementing normalized schemas, row-level isolation, and indexed queries that reduced platform latency from 1.2 seconds to 350 milliseconds.
- Engineered **graceful failure handling for distributed workflows** across 5,000+ monthly jobs using explicit states, validation, reconciliation, and failure isolation, achieving 99%+ reliability while preventing cascading integration failures.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Built **scalable Flask services and React interfaces** supporting 150+ users and 10,000+ daily requests, applying multithreading, persistent results, and fault-tolerant APIs to process 1,000+ files per batch.
- Integrated and critically validated **AI-generated outputs** across multiple LLM and OCR providers, adding usage logs, structured validation, fallback handling, and automated tests to improve correctness while reducing turnaround time 60%.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Developed **reusable API libraries and data components** using OOP patterns, SQL, caching, AWS Glue, and S3, improving response times 30%, pipeline efficiency 68%, storage costs 15%, and maintainability.
- Automated **shared development and deployment frameworks** with GitHub, Jenkins, and 20+ unit/integration tests, generating coverage and failure reports while supporting consistent releases across Windows, Linux, and macOS environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed **broadly shared Node.js services and React experiences** for authentication, records, billing, permissions, and notifications, synchronizing 2M+ records through REST APIs, WebSockets, and Azure SQL across eight product domains.
- Improved **platform reliability, security, and operations** using JWT access, Jest, Selenium, Sentry, Azure Monitor, Docker, and Jenkins CI/CD, cutting deployment time 50% while accelerating diagnosis of unexpected service failures.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built **Spring Boot and Flask platform APIs** for bookings, pricing, payments, and status updates, maintaining sub-200-millisecond responses across 50,000+ records through Firebase synchronization and reliable webhook integrations.
- Collaborated with partners to **design and document reusable interfaces**, mapping 12+ integration points and shipping unit-tested, version-controlled enhancements that improved cross-platform compatibility 15% and project delivery speed 22%.

Skills

Languages: Java, Python, C++, JavaScript, TypeScript, SQL, Bash
Computer Science: Data structures, algorithms, OOP, software design, design patterns, multithreading, distributed systems
Platform: Spring Boot, FastAPI, Flask, Node.js, React.js, REST APIs, microservices, WebSockets, reusable interfaces
Storage & Cloud: PostgreSQL, MySQL, SQL Server, MongoDB, AWS, Azure, Docker, Kubernetes, caching
Engineering: AI-assisted development, Cursor, Claude, testing, documentation, Git, Jenkins, GitHub Actions, CI/CD, Agile/Scrum

Projects

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP*
- Built an **AI-enabled data and experience platform** processing 50,000+ documents with incremental indexing, metadata persistence, secure access, collaborative workflows, and generative summarization, reducing retrieval latency 40% and review time 60%.

Chat2SQL Code | *Java, TypeScript, Oracle, Docker, Maven*
- Enhanced an **AI-powered structured-data application** across Java and TypeScript components, supporting 10+ SQL/NoSQL systems with reusable interfaces, schema exploration, query generation, Docker, Maven, and cross-platform development workflows.

FinNexus Live | *Next.js, Kubernetes, MongoDB*
- Developed a **scalable customer platform** supporting 50,000+ users and 1M+ transactions with secure integrations, real-time updates, MongoDB storage, reusable filtering interfaces, and Kubernetes-based CI/CD automation for reliable operation.